Supplementary Data

Selected complete paired contrasts and feature-level sanity checks

Article type: Supplement

Short title: S2 Data - hypoglycemia dynamics

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S2. Data files

Machine-readable full results are retained in `results/hypoglycemia_t1d_20260723/ndt_paired_contrast_results.csv`, with feature checks in `raw_eeg_feature_paired_results.csv`. Tables below reproduce the prespecified medians used in the main text. Effects are HYPO minus EU.

S2.1 Coarse and stratified coordinates

Metric	N	Delta	p	q
m median	34	−0.7429	0.0002	0.00090
d median	34	0.0278	0.8592	0.9640
e median	34	−0.2440	0.0332	0.0747
m_a median	34	−1.0956	0.0002	0.0018
m_e median	34	−0.6848	0.0004	0.0018
m_o median	34	−0.3578	0.0916	0.1340
d_n median	34	1.0135	0.0008	0.0031
d_l median	34	−0.4404	0.0310	0.0598
d_s median	34	−0.3110	0.2152	0.2421
e_e median	34	−0.7084	0.0176	0.0396
e_s median	34	0.3226	0.1194	0.1612
e_m median	34	1.0242	0.0016	0.0048

Table 1: Paired coordinate medians from the engine-level contrast table.

S2.2 Local flow and reachability

Metric	N	Delta	p	q
3D Jacobian Frobenius median	34	−0.00968	0.2452	0.6654
3D Jacobian rotation median	34	−0.00526	0.4277	0.6947
3D Jacobian trace median	34	−0.00971	0.0378	0.6654
Tube effective dimension median	33	0.2512	0.0124	0.1343
Tube directional entropy median	33	0.0881	0.0298	0.1534
Tube condition number median	33	−2871.5	0.0602	0.1698
Tube log-volume median	33	0.3950	0.2454	0.5367
Tube rotation median	33	0.00437	0.8324	0.9171

Table 2: Local-flow and reachability medians. No reachability endpoint survived the engine-level reachability FDR family.

S2.3 Secondary raw EEG feature checks

Feature	N	Delta	p	q
Theta power	34	0.1297	0.0000022	0.000013
Delta power	34	0.1025	0.000029	0.000086
Alpha power	34	0.1693	0.00433	0.00865
Sample entropy	34	−0.0280	0.0358	0.0430
Permutation entropy	34	−0.0280	0.0358	0.0430
Hjorth complexity	34	0.0846	0.1682	0.1682

Table 3: Secondary raw-feature checks. The exported sample and permutation entropy fields are identical in this release.

Bibliography